

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) FMM — Semester I Examination, 2023-24

Paper No. BCF-116: Business Statistics

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer all questions. All questions carry equal marks (14 marks each).

Question 1. What do you mean by Primary and Secondary Data? What methods are used to collect these? Explain their relative merits and demerits. [14]

OR

Rearrange the following series with equal intervals and then prepare 'less than' and 'more than' cumulative frequency distribution: [14]

oprule Class Interval	Frequency
0-5	3
5-6	2
6-9	7
9-12	5
12-17	16
17-18	12
18-20	15
20-24	20
24-25	8
25-30	10
30-36	2
Total	100
ottomrule	

Question 2. Explain what is meant by 'Central Tendency' and describe the various methods of measuring it. [14]

OR

Calculate the mean and median from the following series: [14]

oprule Class Interval	Frequency
0-9	32
10-19	36
20-29	20
30-39	30
40-59	48
60-79	24
80-99	2
ottomrule	

Question 3. What do you understand by Skewness? Point out its role in analyzing a frequency distribution. [14]

OR

Calculate the standard deviation and its coefficient from the following data: [14]

oprule Marks	Frequency
10–20	6
20–30	8
30–40	15
40–50	12
50–60	6
60–70	2
70–80	1
ottomrule	

Question 4. What are the components of a time series? How would you find out the trend values in a time series by the method of least squares? [14]

OR

Find out the value of the trend by three-yearly moving averages and plot them on a graph paper using the following data: [14]

oprule Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Value	10	15	12	18	15	22	19	24	20	26	22	30	25
ottomrule													

Question 5. How are index numbers constructed? Discuss the importance of the choice of base year and selection of weights in the construction of cost of living Index Numbers. [14]

OR

Calculate Fisher's Ideal Index number from the data given below: [14]

oprule Item	Base Price (p_0)	Base Quantity (q_0)	Current Price (p_1)	Current Quantity (q_1)
A	6	50	10	60
B	2	100	2	120
C	4	60	6	60
D	12	40	15	25
E	8	50	12	35
ottomrule				

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